

SORAIDA QUINTERO

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SUMMARY

Ph.D. Candidate with experience managing projects and applying statistical analysis, predictive analytics, machine learning, and simulation modeling to understand complex social, environmental, and urban systems. Experienced in integrating, validating, and analyzing large and diverse datasets to identify patterns, evaluate outcomes, and support evidence-based recommendations. Skilled in data management, statistical modeling, spatial analysis, project coordination, and communicating analytical findings to technical and non-technical audiences. Prior experience in the mortgage and real estate industries provides additional expertise working with consumer financial information, housing market trends, regulatory requirements, and data-informed problem solving within highly regulated environments.

TECHNICAL SKILLS

Programming & Data	Analytics & Modeling
R	Statistical Analysis
Python	Predictive Analytics
SQL	Machine Learning
Bash	Spatial Statistics
Linux/Unix	Agent-Based Modeling
GitHub	Simulation Modeling
Jupyter Notebook	
Additional Skills	
Data Management	Data Integration
Data Validation	Microsoft Word
Data Cleaning	PowerPoint
Data Wrangling	Microsoft Excel
Reproducible Workflows	Spanish (Fluent)

RESEARCH EXPERIENCE

- Graduate Research Assistant** | University of Illinois Chicago | 2021–Present
- Design and evaluate statistical, machine learning, and agent-based modeling approaches to examine social and ecological systems.
 - Integrate, clean, validate, and analyze data from surveys, ecological observations, spatial datasets, and generated outputs.
 - Develop predictive and statistical models to identify behavioral, environmental, and spatial patterns.
 - Translate analytical findings into reports, presentations, and research products for interdisciplinary audiences.
- Teaching Assistant** – Biostatistics with R | University of Illinois Chicago
- Support students in applying statistical methods, data analysis, and R programming.
 - Assist with hypothesis testing, regression analysis, experimental design, and interpretation of analytical results.
- Graduate Fellow** | Purdue University Policy Research Institute | 2018–2021
- Managed a Mellon Foundation-funded interdisciplinary initiative involving five grant-supported projects across multiple academic departments.
 - Established timelines, monitored progress, and ensured completion of grant deliverables.
 - Prepared reports and project documentation for university leadership and foundation reporting requirements.

PROFESSIONAL EXPERIENCE

Licensed Managing Real Estate Broker | 2003–Present

- Licensed real estate professional since 2003 with experience analyzing housing market trends, neighborhood characteristics, property values, and local market conditions.
- Interpret transaction data, pricing patterns, inventory levels, and market indicators to support client recommendations.
- Evaluate market trends, risks, and opportunities to assist purchasing, selling, and investment decisions.
- Coordinate transactions while ensuring compliance with state and federal regulations.

Mortgage Broker | 1998–2010

- Evaluated consumer financial records, credit reports, income documentation, debt-to-income ratios, and lending criteria.
 - Assessed risk, loan eligibility, and financing options using information from multiple financial sources.
 - Reviewed and validated financial documentation to ensure compliance with lending requirements.
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SELECTED PROJECTS

Neighborhood Gentrification Predictions through Real Estate Market Analysis

Created a property analysis model to predict neighborhood gentrification patterns using property and area characteristics with publicly available housing and census data.

Methods: Regression and clustering in R using tidyverse, ggplot2, and machine learning algorithms (random forest, gradient boosting).

Outcome: (preliminary) Improved property value prediction accuracy and identified socio-economic indicators driving neighborhood change.

Spatial Mimicry in Chicago Residential Landscapes

Examined how social interactions and neighborhood factors shape spatial patterns of residential yard management.

Methods: Applied spatial statistics and logistic regression to assess plant clustering patterns by measuring spatial autocorrelation using Moran's I and Mantel r tests.

Outcome: Demonstrated that social behaviors at the neighborhood level are spatially autocorrelated, revealing mimicry patterns among adjacent households.

Free-Roaming Dog Population Modeling

Developed agent-based models to simulate dog population dynamics, and feeding behaviors across diverse ecological and cultural contexts.

Methods: Implemented demographic and movement modeling, intervention simulations (vaccination, sterilization, cultural), and spatial analyses

Outcome: (preliminary) Identified context-specific drivers of population level change, and demonstrated that targeted interventions for free-roaming dogs yield significantly improved management outcomes.

PUBLICATIONS

Quintero S, Ballut N, Johnson P, Minor E, (2026). Examining the mechanisms that create spatial patterns in vegetation across residential landscapes. *Computers Environment and Urban Systems*. Manuscript submitted.

Quintero S, Zollner P, Atallah S, (2026). Understanding the Disturbance of Human Recreation on Wildlife using Multiple Dynamic Agents within an IBM Framework. *Ecological Modelling*. Manuscript in progress.

Quintero S, Weldon L, Clawson R, Roberts A, (2026). Engaging Communication for High Impact Research: Four models for organizing interdisciplinary research teams. *Proceedings of the National Academy of Sciences*. Manuscript in progress.

EDUCATION

University of Illinois Chicago - PhD, Ecology and Evolution 2026

Purdue University - MS, Forestry and Natural Resources 2021

Chicago State University –BS, Environmental Science 2016